Synthesized Intelligence: Integrating Morphological Computation for Co-evolving Adaptive Robotic Systems

MSCD Pre-thesis I

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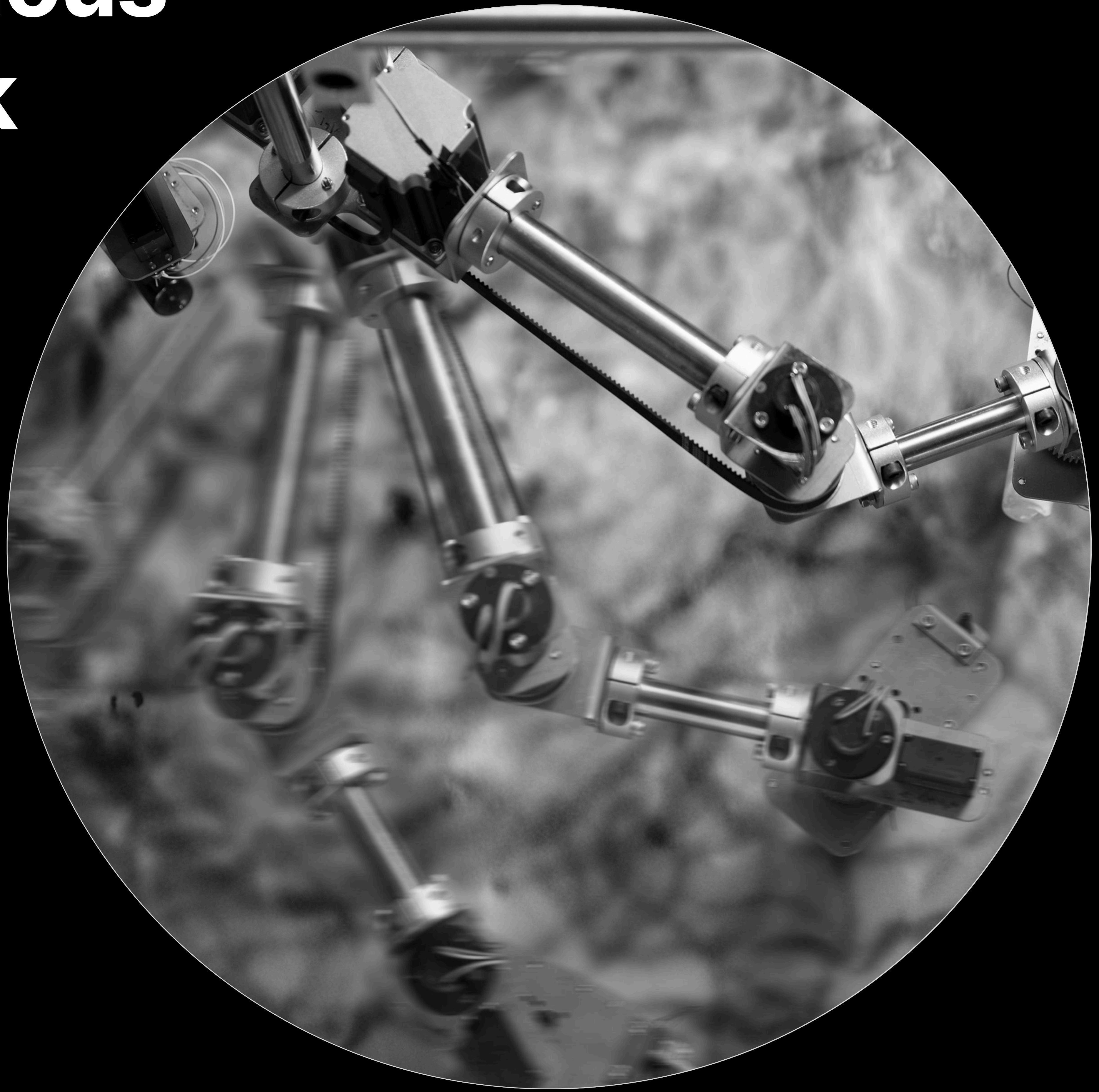
Research Question

How can embodied or **'morphological' computation**—combined with **bio-inspired generative algorithms** and **machine learning**—enable robotic systems to **adapt their forms and behaviors** in real time, thereby **reducing reliance on traditional top-down contro**l methods?

Problem Statement

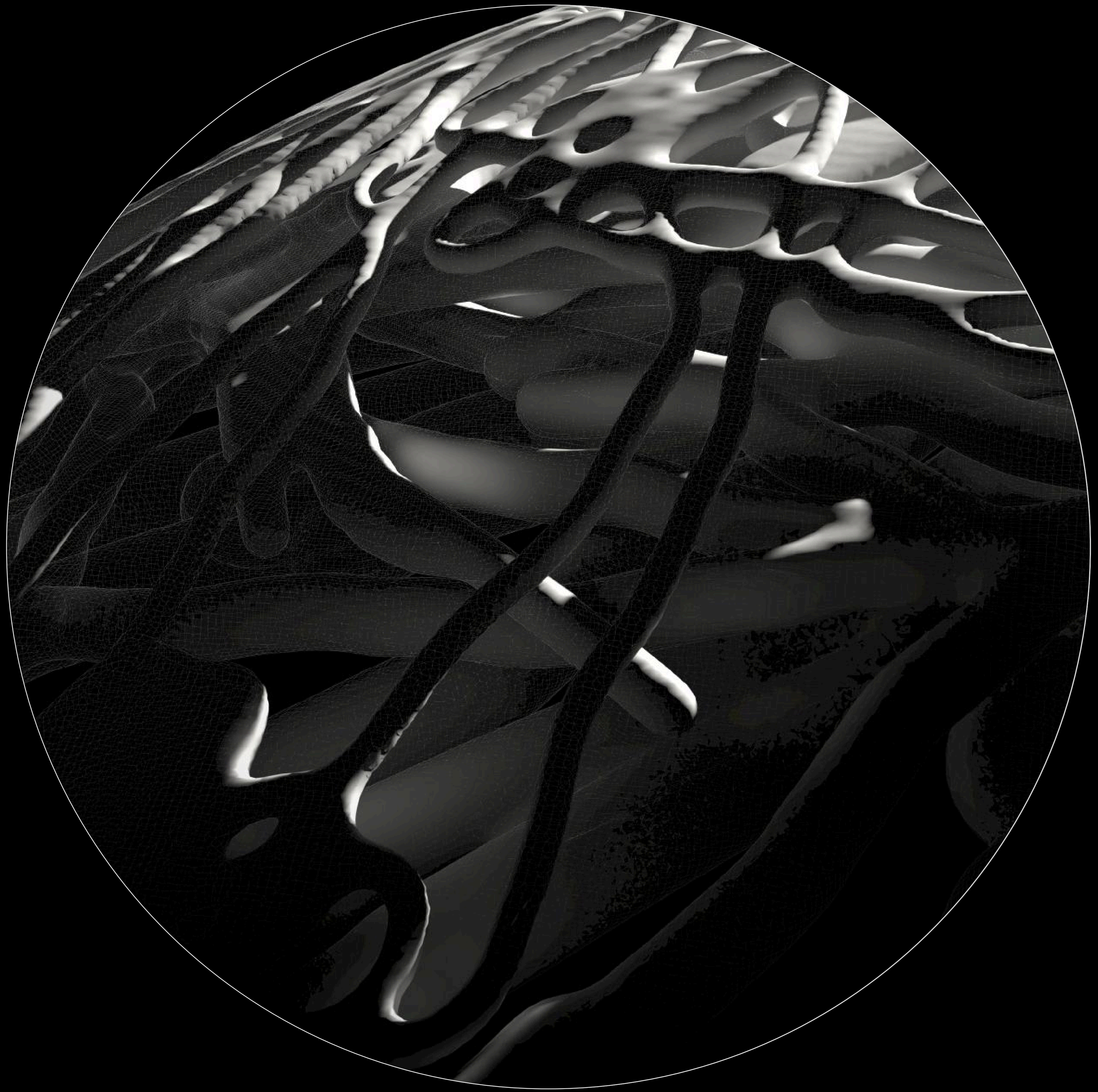
Traditional robotic systems often **separate physical design from control algorithms**, leading to rigid structures that lack adaptability. This separation overlooks the potential of morphological computation, where physical structures contribute to computational processes. **By integrating** bio-inspired generative algorithms with machine learning, there's an opportunity to develop robotic systems that adapt both their form and function in response to environmental stimuli, enhancing resilience and efficiency.

Previous Work



Drawing Machine

11/2023



Slime Mold Structure

11/2024

Context

Morphological Computation

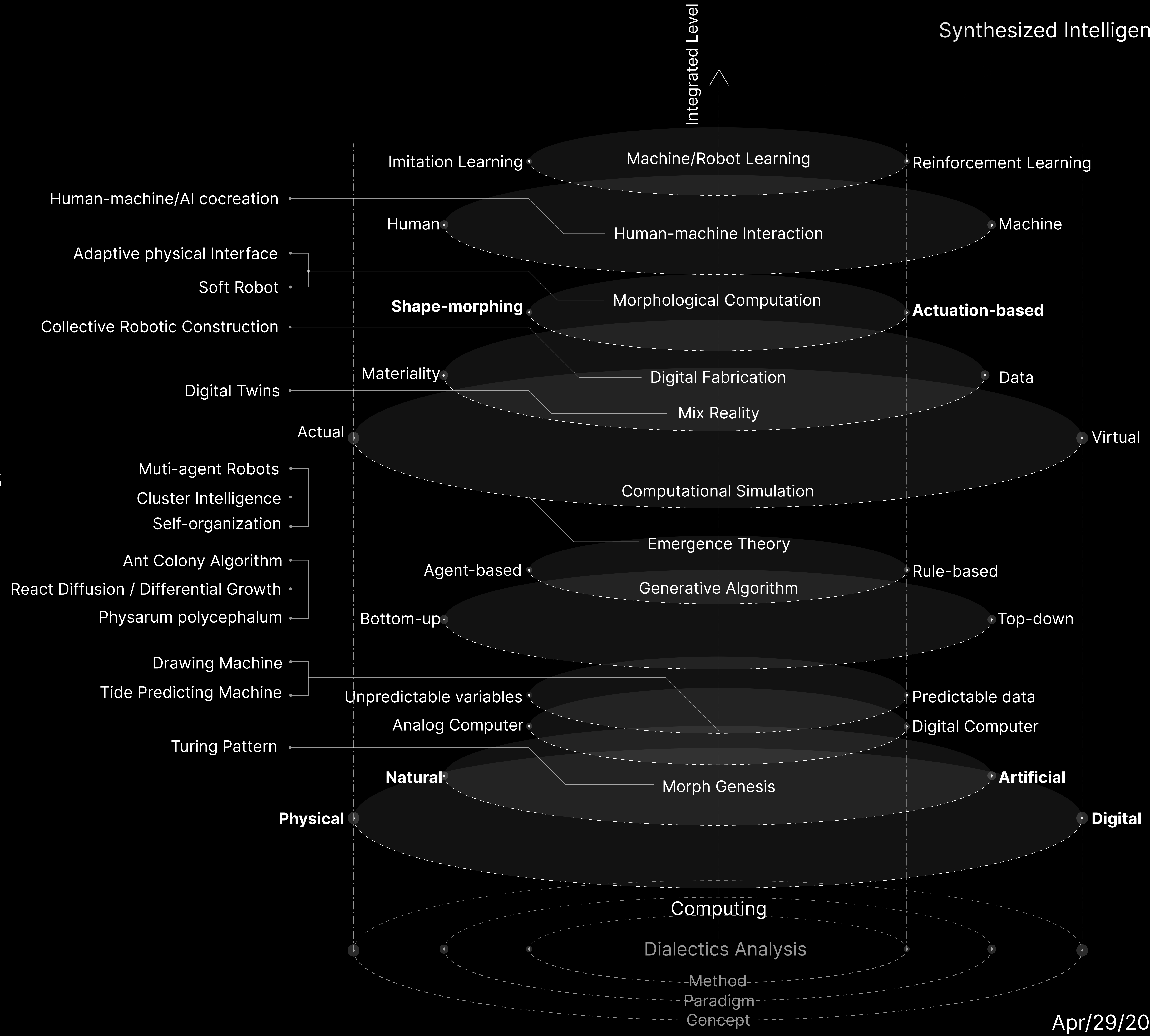
Emerging discipline studying how physical structure or material properties can perform a portion of what is typically called “computation.”

Generative / Bio-inspired Algorithms

Techniques like differential growth, reaction-diffusion, or cellular automata that replicate natural processes of form-making

Machine Learning and Robotics

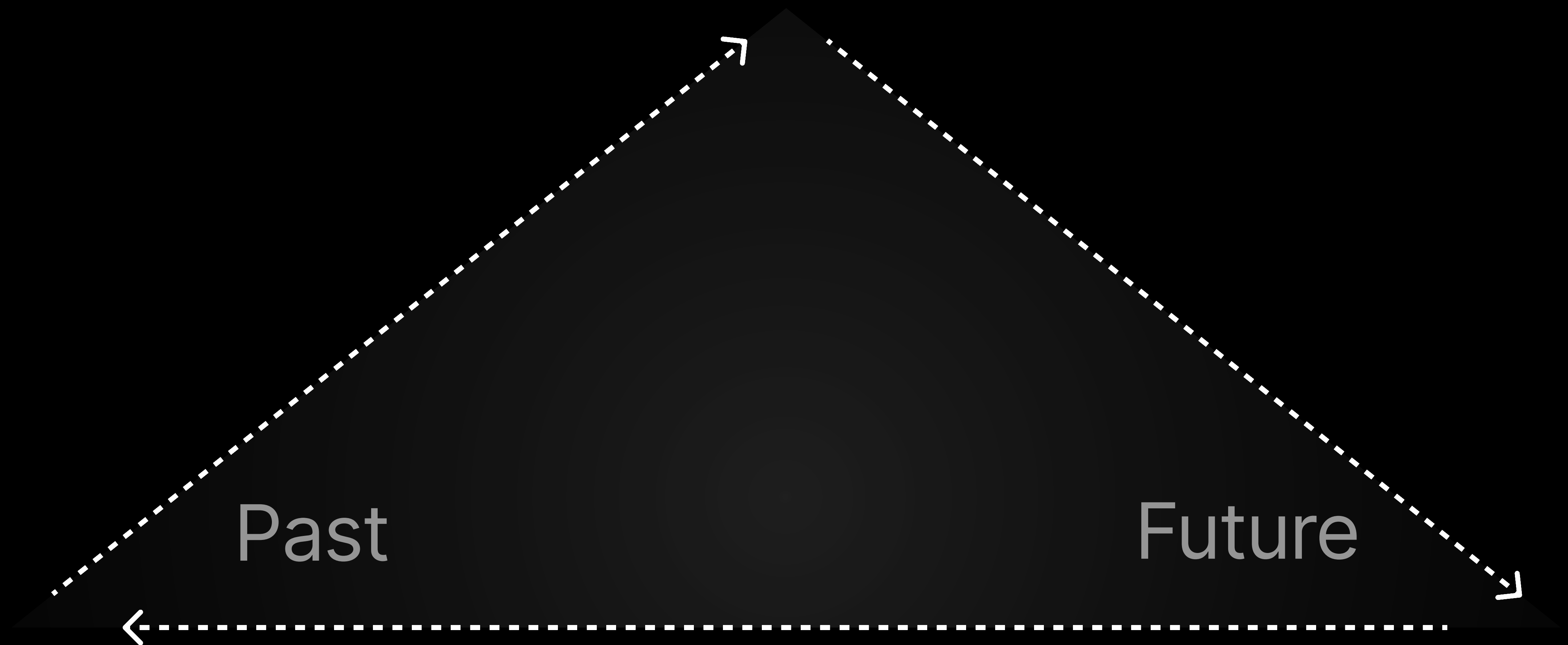
The link between shape and control strategies is traditionally an afterthought—shape is fixed, and software does the rest. Instead, this project foregrounds adaptive shape or structure as a vital part of intelligence.



Trinity of the Integration

Physical

Morphological Computation



Digital

Bio-inspired Generative Algorithms

Intelligence

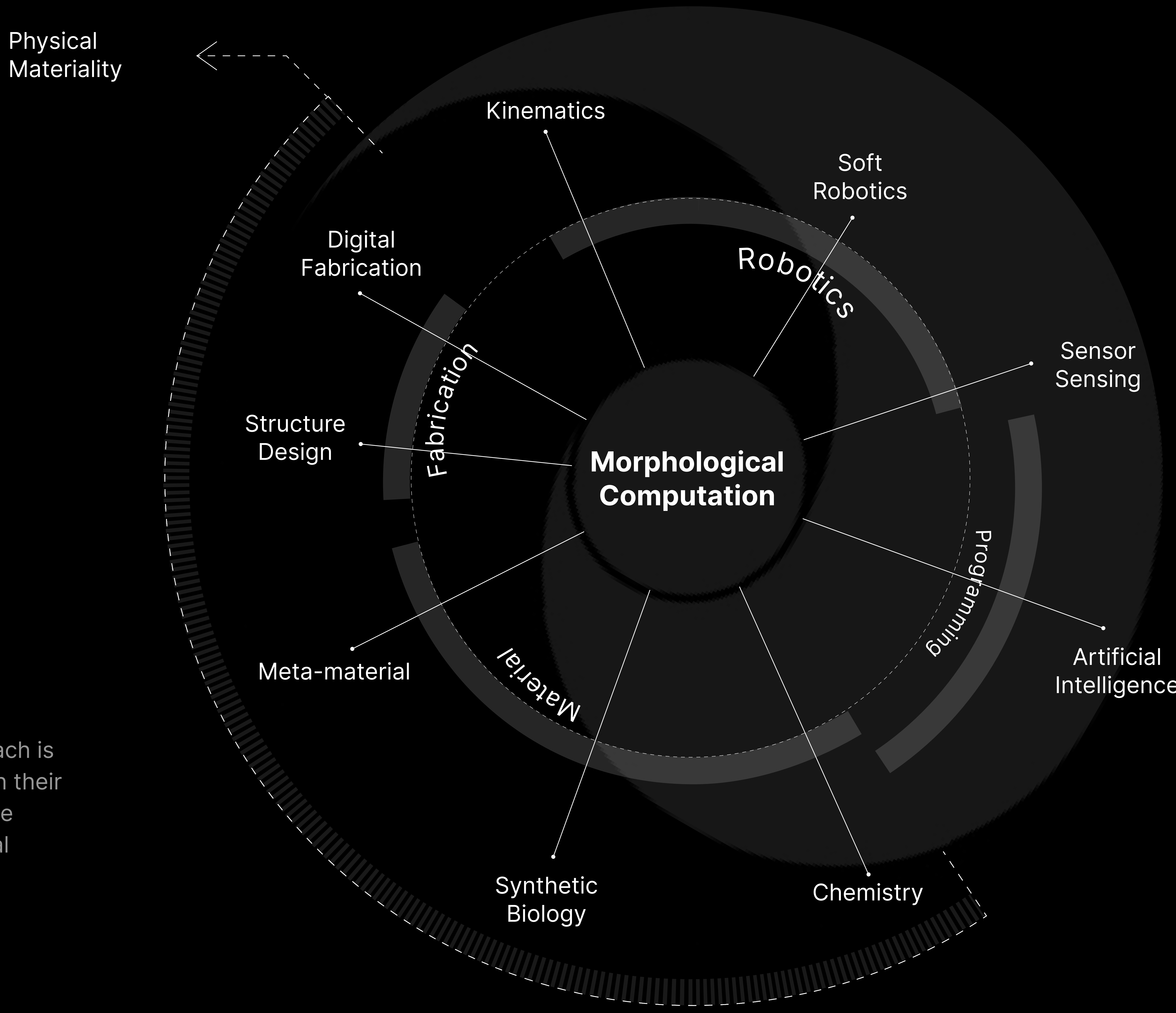
Machine Learning and Human Factor

Definition

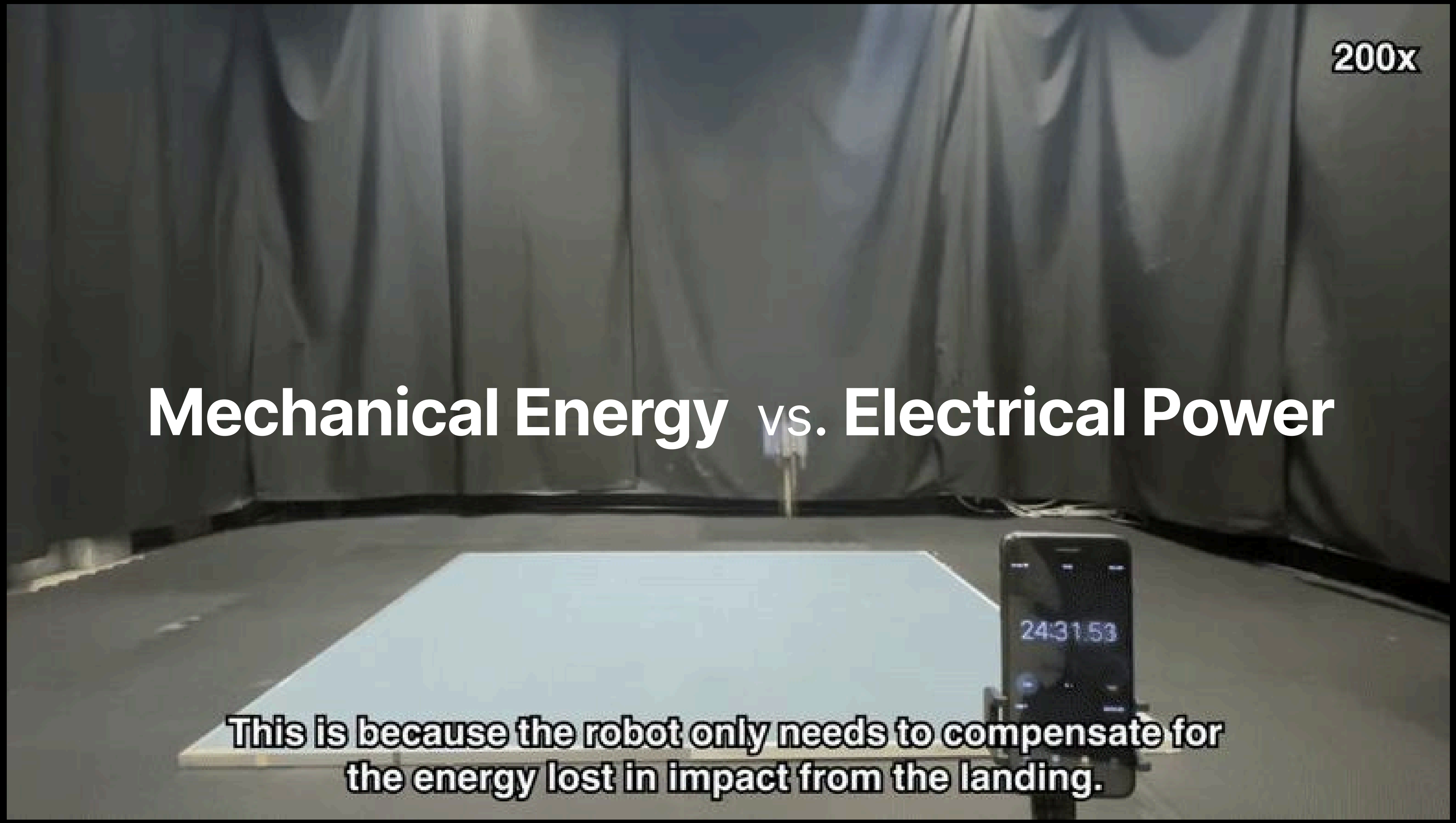
Morphological Computation

 # Programmable Material
 # Computing is Physical
 # Multimodal Locomotion
 # Soft Robotics

Morphological computation is a design approach used in robotics that proposes the exploitation of morphological features in order to build intelligent robotic systems and improve their performance. This approach is inspired by observations of biological systems, which all rely heavily on their body morphologies. Taking into account the material properties and the corresponding dynamics of the physical body of a robot, morphological computation aims to improve sensing, control, and learning.



Gap Analysis - Compared to the Conventional System



Adaptability

Traditional systems need complex sensor+actuator stacks to conform to terrain; these seed carriers adapt “for free,” purely through their shape. [1]

Efficiency

Conventional bipeds/quads burn kilowatts to hop or flap; the hybrid hopper re-uses mechanical energy. [2]

[1] Luo, D., Maheshwari, A., Danielescu, A. et al. Autonomous self-burying seed carriers for aerial seeding. Nature 614, 463–470 (2023). https://doi.org/10.1038/s41586-022-05656-3
[2] Bai, Songnan & Pan, Qiqi & Ding, Runze & Jia, Huaiyuan & Yang, Zhengbao & Chirarattananon, Pakpong. (2024). An agile monopedal hopping quadcopter with synergistic hybrid locomotion. Science robotics. 9. eadi8912. 10.1126/scirobotics.adi8912.

Gap Analysis - Problem Space

Gap 1

Adaptive shape + control

No existing system simultaneously adapts both physical geometry and control policy on the fly in a real-world robotic platform.

Gap 2

Closed loop

Generative-morphology pipelines are seldom paired with sim-to-real reinforcement-learning to close the loop.

Gap 3

Metrics

Performance metrics focus on task success, not on how morphology drives energy or resilience gains.

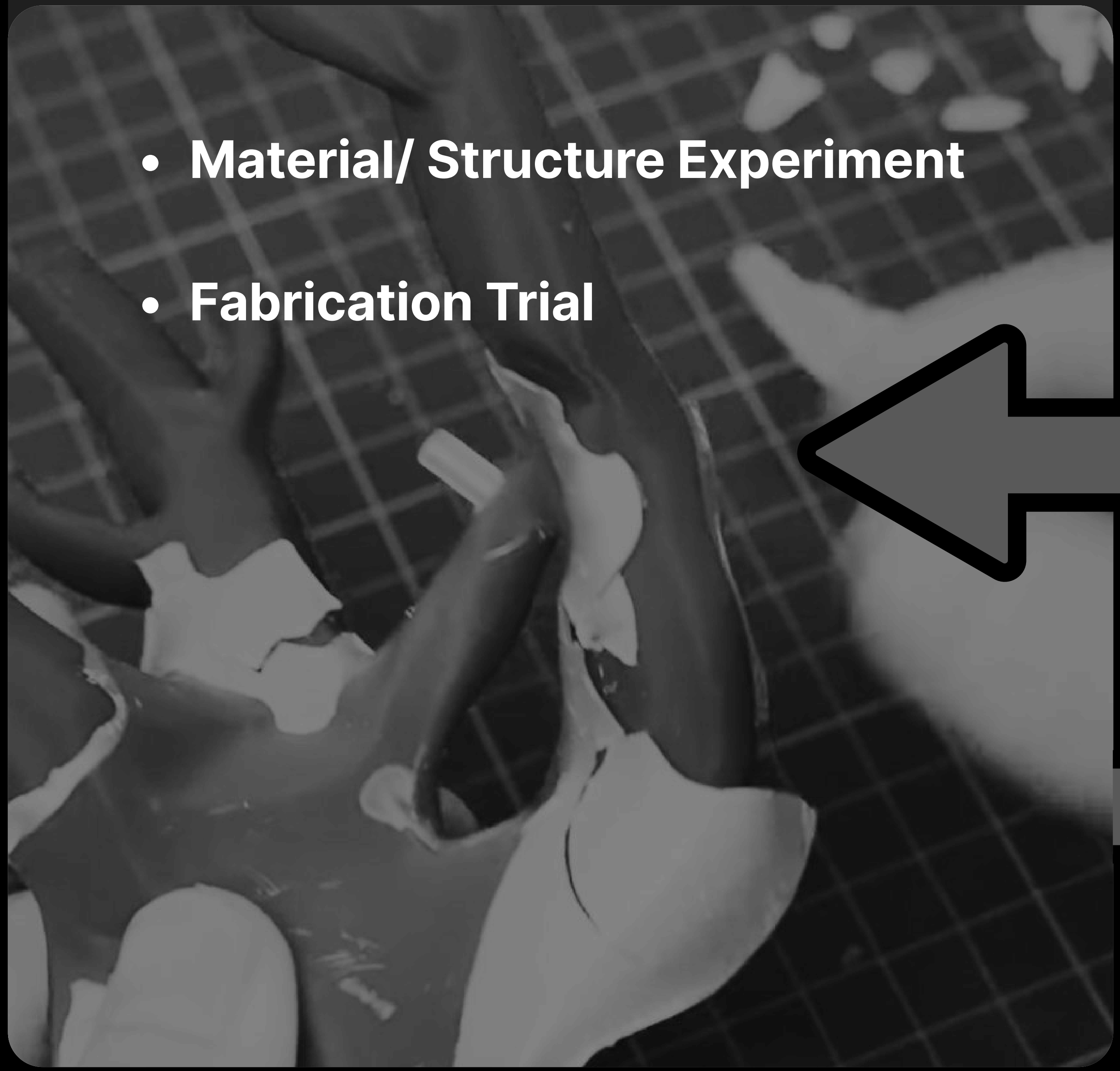
Gap 4

Human factor

Lacking of human intervention or integration with human factor

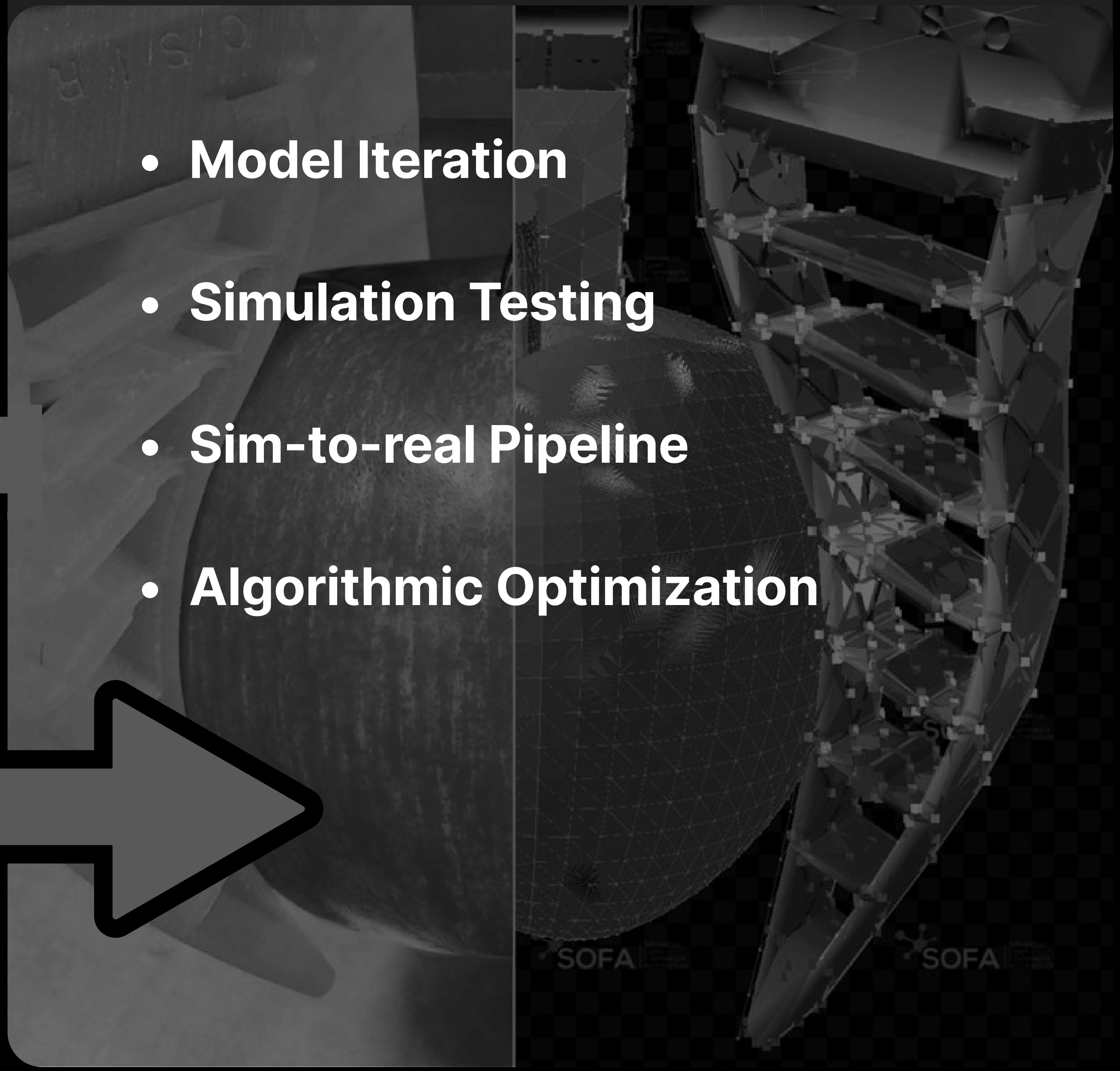
Game Plan

Phase 1 Physical Prototyping



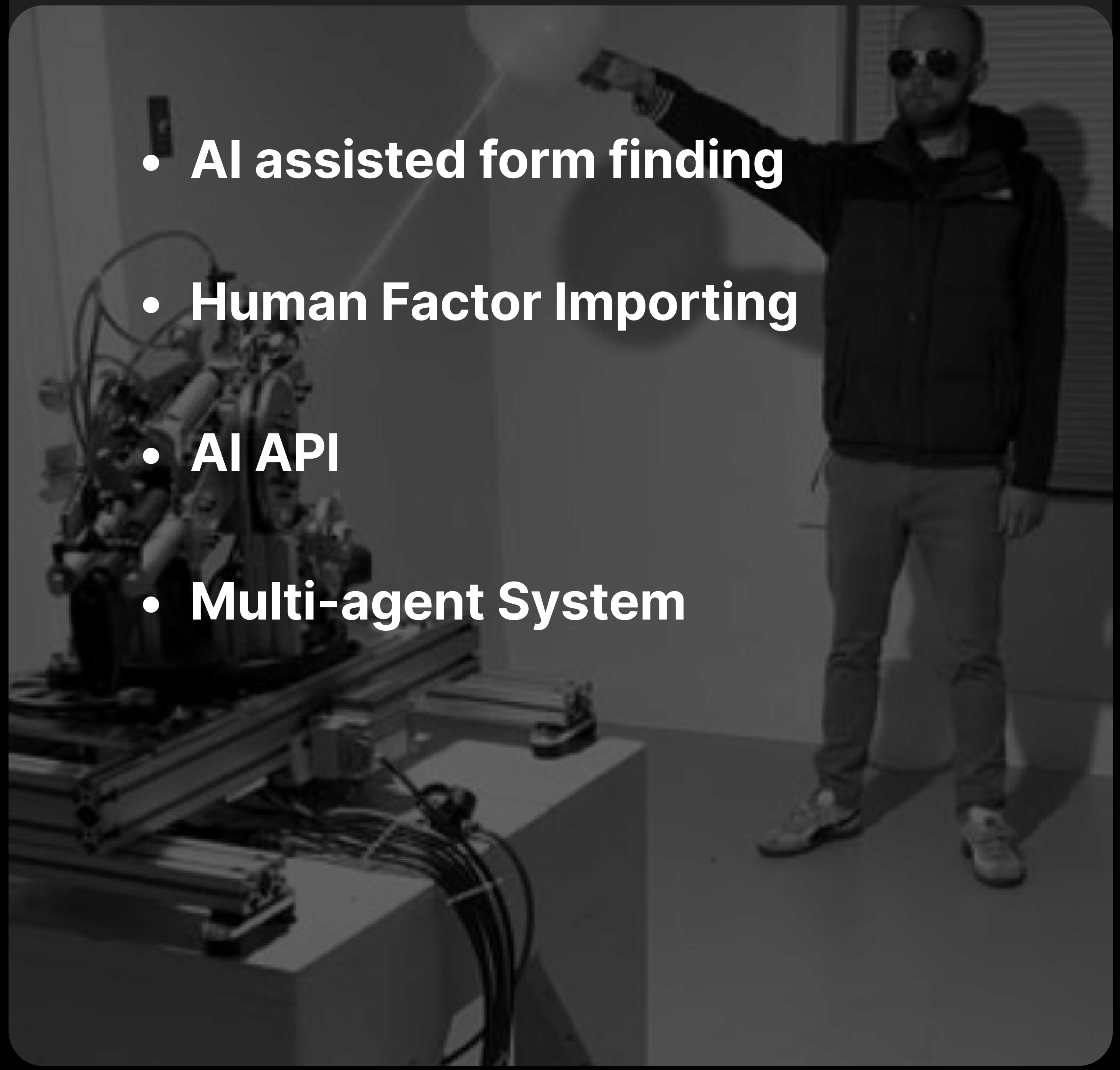
- Material/ Structure Experiment
- Fabrication Trial

Phase 2 Digital Synthesizing

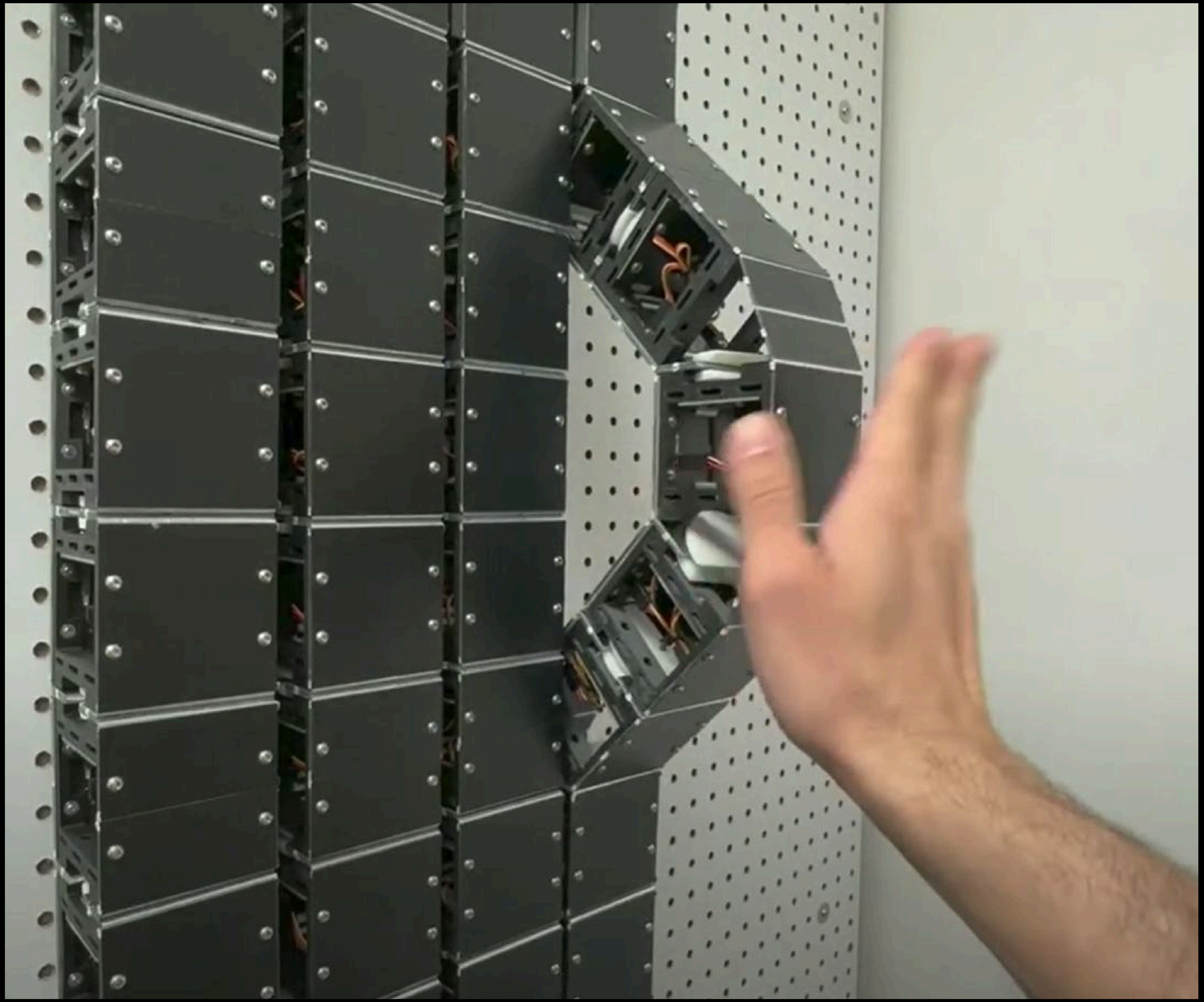


- Model Iteration
- Simulation Testing
- Sim-to-real Pipeline
- Algorithmic Optimization

Phase 3 Intelligence Embedding



- AI assisted form finding
- Human Factor Importing
- AI API
- Multi-agent System



Time Line

2 - Sim-to-real Loop

Jul

Material/structure
experiments finished

Sep

start sim-to-real loop

Feb

Prototype v1

2024

2025

May

Kick off
Work as RA

Aug

Fabrication trial #1

Oct

Closed-loop

Nov

Fabrication trial #2

Apr

Full system demo

1 - Material

3 - Final Demo

Suggestion Seeking

A. Morphology focus

- Option 1: single “hero” craft (go deep)
- Option 2: compare 2–3 crafts (breadth)

Trade-offs I see: depth vs generalisability.

B. Evaluation lens

- Energy-use / resilience metric OR task-success only?
- How would you measure “intuitive thinking” in a robot?